

 accenture



# Top 45 Accenture Data Engineer Interview Questions

For professionals with **2–5 years of experience**



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Data Engineer

# Data Engineering Fundamentals

1. Key responsibilities of a Data Engineer
2. OLTP vs OLAP systems
3. ETL vs ELT – use cases
4. Common data engineering challenges
5. Data partitioning in warehouses



# SQL & Performance Tuning

- 6. How to optimize a slow SQL query
- 7. Indexing – what & why
- 8. Types of SQL joins with examples
- 9. Denormalization – purpose & use
- 10. Handling duplicates in SQL



# Apache Spark Core Concepts

11. Spark vs Hadoop MapReduce

12. RDD vs DataFrame vs Dataset

13. cache() vs persist()

14. Shuffle – what it is & how to optimize

15. Types of transformations in Spark





# Spark Coding Challenges (PySpark/Scala)

16. Remove duplicates from a DataFrame

17. Calculate moving average

18. Scala program – WordCount using Spark

19. Fill nulls with column mean

20. Group by + sum in PySpark



# Hive Data Warehouse

21. Partitioning vs Bucketing

22. Hive internal storage

23. File formats in Hive (Parquet, ORC, etc.)

24. Dynamic partitioning – with example

25. Hive indexing



# Apache Airflow (Workflow Management)

26. What is Airflow?

27. Pipeline scheduling in Airflow

28. DAGs in Airflow

29. Handling failed tasks

30. What is backfilling?



# Cloud & Storage (AWS & Hadoop)

- 31. AWS S3 vs HDFS
- 32. How AWS Glue works
- 33. Using AWS Lambda for ETL
- 34. Redshift vs Athena
- 35. S3 optimization for queries





# Kafka & Real-Time Processing

36. What is Kafka & how it works

37. Producer vs Consumer

38. Ensuring message ordering

39. Topics & partitions

40. Fault tolerance in Kafka



# Optimization Techniques

41. Spark performance tuning techniques

42. Data lake storage optimization

43. Role of Parquet/ORC in performance

44. Vectorization in Pandas & Spark

45. Best practices for SQL on big data



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